

Tarcísio G. Rodrigues

Senior Software Engineer

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Overview

Senior backend engineer with 10+ years building low-latency, high-reliability systems in fintech (payments, market data, risk analysis). Currently focused on AI-native engineering workflows: customizing coding, review, and dispatch agents in spec-driven development frameworks, and contributing upstream to open-source agent tooling.

Employment

- 2026 **Software Engineer**, *Cetacea*, São Paulo, Remote
- Designed and developed .NET (C#) backend services for the customer ticket lifecycle, plus middleware that interfaces with the Matrix communication service.
 - Built and maintained a React/TypeScript/GraphQL frontend layer, adding a dedicated real-time widget to speed up customer onboarding.
 - Developed an ASP.NET Core backend that consolidates onboarding workflow metrics and exposes them through a NestJS BFF.
 - Led adoption of spec-driven development (SDD), authoring the initial spec templates and establishing the review workflow.
 - Customized the coding, review, and dispatch agents in two SDD frameworks (acai.sh and gsd-core), tuning prompts, agent roles, and workflows to the team's codebase and conventions.
 - Made the setup work out of the box with both Claude Code and OpenCode, providing a tool-agnostic paved road for AI-assisted development.
- 2025–2026 **Software Engineer**, *Mottu*, São Paulo, Hybrid
- Built .NET (C#) backend services to integrate with both Santander's PIX payment infrastructure and BBVA México's payment services.
- 2024–2025 **Software Engineer**, *mParticle*, New York, Remote
- Contributed to the evolution and support of a distributed querying system for extracting marketing performance analytics (audience segmentation, funnel analysis) from large-scale customer event data. Primarily developed backend services in Java, with additional contributions to the JavaScript frontend.
- 2016–2023 **Software Engineer**, *XP Inc.*, São Paulo
- Implemented market data feed software in C++ with Boost (Asio, Signals2, Multi-Index, ICL), QuickFIX, QuickFAST and SBE reference implementations. Worked with market data protocols, FIX/FAST and SBE encodings, and B3's UMDF platform. Designed and developed software infrastructure to integrate backend and frontend services, employing .NET microservices, Kafka, MongoDB, Docker, Kubernetes and WebSockets.
- 2014–2016 **Software Engineer**, *FIT – Institute of Technology*, Sorocaba
- Developed a C++ library for payment applications. Wrote CMake and Python scripts for build automation. Worked with the EMV standard and Verifone's SDK.
- 2014 **Software Engineer**, *MAPS – Financial Market Solutions*, São Paulo
- Developed and maintained risk analysis software in C++ with Microsoft Foundation Classes. Maintained and extended the Jenkins CI system.

- 2013–2014 **Software Engineer**, *IGS – International Gaming Solutions*, São Paulo
 Developed an embedded backend for land-based bingo games with Boost. Ported legacy C code to Linux. Developed a continuous integration system on Buildbot. Worked with IGT's SAS (Slot Accounting System) protocol. Led a proof of concept for an internal (on-premises) Django web application for managing and consolidating slot accounting.
- 2012–2013 **Software Analyst**, *FiberWork – Optical Communications*, Campinas
 Developed Android applications and embedded Java software for remote testing units of passive optical networks.
- 2011 **Adjunct Professor**, *EEP – Piracicaba Engineering School*, Piracicaba
 Taught Analytic Geometry and Linear Algebra.

Education

- 2008–2010 **MSc in Philosophy**, *University of Campinas (Unicamp)*, Campinas
 In the area of Mathematical Logic, with a dissertation on logic programming.
- 2002–2007 **BEng in Computer Engineering**, *Unicamp*, Campinas

Technical highlights

- AI tooling Claude Code, OpenCode, spec-driven development (acai.sh, gsd-core), agent customization (coding, review, dispatch agents), prompt and context engineering
- Languages C#/.NET, C++, Java, Python, TypeScript/Node.js
- Backend ASP.NET Core, NestJS, gRPC, REST, GraphQL, Boost/Asio
- Distributed Kafka, RabbitMQ, LMAX Disruptor, io_uring, libuv
- Data PostgreSQL, SQL Server, MongoDB, Redis
- Observability Datadog, Grafana, Prometheus, distributed tracing, APM
- Infra & CI/CD Docker, Kubernetes, ArgoCD, Nix/NixOS, GitHub Actions, Jenkins, GCP, AWS
- Security OAuth, JWT
- Fintech PIX payment APIs, FIX/FAST/SBE, B3's UMDF platform, EMV
- Practices DDD, CQRS, TDD, Clean Code, Scrum, Kanban

Languages

- English Advanced/Fluent.
- Spanish Conversational.
- Portuguese Native.